Getting familiar with the Praat environment

Once launched, two windows are opened in Praat. One is the *Object* window and the other is the *Picture* window. We will discuss only the *Object* window, since most of the basic functionalities can be achieved through this interface alone. Praat is special in that it offers dynamic buttons upon selection of different sound objects. We will quickly go through the important buttons and their features. Readers are encouraged to explore the features of the rest of the buttons themselves.

Menu	Functionality
Praat	New and open Praat script (deals with creation and opening of Praat scripts)
New	Records sounds with different options and creates a text grid for annotations
Open	Deals with reading different sound files and loading them into *Object* window
Save	Deals with saving the recorded and processed sound objects into the desired file types

Table 2: Praat main menu showing key functionalities

It is very easy to record a sound using the *New >> Record Mono Sound* menu option. Selecting this option presents a window that is ready to record the sound with some default set parameters. You can modify sampling frequency or channel types. If you do not understand the terms, then keep the default settings unchanged. Pressing the *Record* button starts recording until the *Stop* button is pressed. You can test your recording by clicking the *Play* button. If the recording is acceptable, then specify some meaningful name to the recorded sound file and click on *Save* to list and then on *Close*. The sound object will now appear in the *Object* list. If the recording is not satisfactory, then repeat the recording. A typical screenshot of the Praat Sound Recorder is shown in Figure 5. Note that recording the sound and saving it saves the sound file as the object in the current session only. If you want to permanently save the recorded sound, then use *Save >> Save as WAV file…* menu.

Various functionalities

o *Reverse a sound:* To record or read a sound in the *Objects* window, select the sound object which you want to reverse. Click on *Modify >> Reverse*, and play the sound once again. It will play the sound in reverse. To save the reverse version of the sound, use the *Save* menu from the top panel. Performing the same operation again will get you back to the original sound.

o *Text-to-speech synthesis:* You can create a speech synthesiser from a given text. Click on *New >> Sound >> Create SpeechSynthesizer.* Select the language and voice variant (male/female) and click on the *Play Text* button.

o *Amplify a sound:* A soft sound can be made louder by scaling its amplitude. To do this, select the target sound object, and click on *Modify >> Formula*. In the window that appears, enter the scaling factor for the sound. For example, *self*2* will double the amplitude of the sound.

Figure 5: The Praat sound recorder

Scripting in Praat

A script comprises a set of commands that need to be executed in a specific sequence to perform a certain task. Praat also supports scripting. Users can write scripts, which are interpreted by an interpreter.

Executing scripts

Let us start with a simple script. Click on *Praat >> New Praat Script* to open the script editor. The following are some of the usual commands that are often needed in scripts.

Command	Task
Play	Plays a selected sound object
Remove	Removes a selected sound object from the *Object* window
writeInfoLine:	Prints the message passed as the argument
appendInfoLine:	Appends the message into the existing display rather than overwriting
Clearinfo	Clears the Praat *Info* window, where the output is displayed
runScript:	Executes an external script

Table 3: Some common scripting commands

Dealing with forms

Praat facilitates the use of forms to get input from users at runtime. Let us look at the use of forms with an illustrative example.

Example 1: To get personal details from the student (such as name, age, gender and results) through the form, and display these details, type: